

MCMC HOMEWORK - DERIVATIONS

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IMPORTANCE SAMPLING

$$1 = \int_0^2 x^{\frac{1}{3}} e^{-x^2} dx$$

2. Monte Carlo mean = 0.67159 $\hat{\mu}_{MC}$
SD = 0.00043

3. Importance sampling

$$\int_0^2 c x^{\frac{1}{3}} dx = 1$$

$$c \cdot \frac{3}{4} \cdot x^{\frac{4}{3}} \Big|_0^2 = c \cdot \frac{3}{4} \cdot 2^{\frac{4}{3}} \rightarrow c = \frac{\sqrt[3]{4}}{3}$$

$$F(x) = \int_0^x c t^{\frac{1}{3}} dt = c \cdot \frac{3}{4} x^{\frac{4}{3}} = \frac{4^{\frac{1}{3}}}{3} \cdot \frac{3}{4} \cdot x^{\frac{4}{3}} = \left(\frac{x}{2}\right)^{\frac{4}{3}}$$

$$F^{-1}(x) = 2x^{\frac{3}{4}} : [0, 1] \rightarrow [0, 2]$$

mean = 0.67131 $\hat{\mu}_{IS}$
SD = 0.00028

4. Variance reduced by half.

Both relatively stable and converge in 10^6 iter.

Importance sampling is more efficient (factors cancel out).

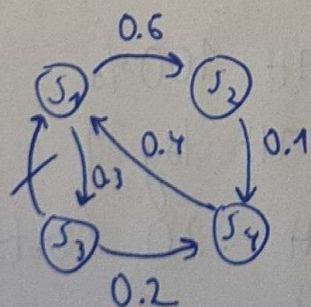
$g(x)$ is not the most suitable as $x \rightarrow 2$

Proposal distribution should match the shape of $f(x)$.

MARKOV CHAINS

1. Irreducible

$G(K)$:



all states communicate

2. $\pi \approx [0.14 \quad 0.37 \quad 0.35 \quad 0.11]$

3. Ergodic theorem

$$\frac{N_i(n)}{n} \xrightarrow{\text{a.s.}} \pi_i$$

$$\text{SLLN: } \frac{1}{N_i(n)} \sum T_i \xrightarrow{\text{a.s.}} E[T_i] \quad \text{and} \quad n = \sum T_i$$

$$\pi_i = \frac{1}{E[T_i]}$$

$$4. \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n g(X_t) \stackrel{\text{a.s.}}{=} E_{\pi}[g(x)] = \sum_{i=1}^4 \pi_i g(i)$$

5. Aperiodic because it contains self loops, for example $K_{ii} > 0$.

Therefore $\gcd(\text{cycle lengths}) = 1$

METROPOLIS - HASTINGS

1. Posterior density

$$\begin{aligned} p(\lambda|x) &= p(x|\lambda) \cdot \pi(\lambda) \propto \lambda^{\sum x_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\beta\lambda} \\ &\propto \lambda^{\alpha-1+\sum x_i} e^{-(\beta+n)\lambda} \sim \text{Gamma}(\alpha+\sum x_i, \beta+n) \end{aligned}$$

2. Acceptance probability

$$A(\lambda, \lambda') = \min\left(1, \left(\frac{\lambda'}{\lambda}\right)^{\alpha+\sum x_i} e^{-(\beta+n)(\lambda'-\lambda)}\right),$$

where $\lambda' = \lambda \cdot e^{\varepsilon}$ and $\varepsilon \sim N(0, \sigma^2)$ "q($\lambda'|\lambda$)"

sigma square = 0.65

acceptance rate = 33%

3. ESS ≈ 50

$$4. q(\lambda') = \text{Gamma}(20, \frac{1}{4})$$

Acceptance rates: RWMH 32%, IMH 100%

Autocorrelation: RWMH lag ~ 10 , IMH lag 1

ESS post burn-in: RWMH 210, IMH 330 (total)

Overall the proposal distribution matches the posterior, which gives us a perfect importance sampler.